INDEX

S. No.	Date	Name of the Experiment	Page No.	Marks Awarded	Signature
01	21.01.26	verification of kirchoff's current law and kirchoff's voltage law		99	
02	28.01.26	open circuit and load characteristics of DC Shunt generator		99	
03	18.2.2026	Staircase wiring		100	
04	18.2.2026	Half wave and full wave rectifier		100	
05	25.2.26	zener diode as voltage regulation		100	
06	11/3/26	characteristic of LVDT		100	

Completed
11/03/26

Expt. No. 6

Characterstic of LVDT
(Characterstic variable d)

AIM:

To understand and simulate the relation between core displacement and output Voltage

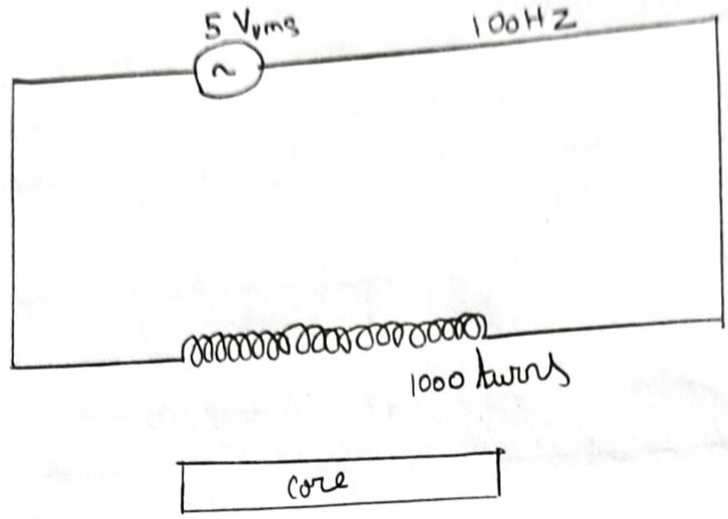
Apparatus Required

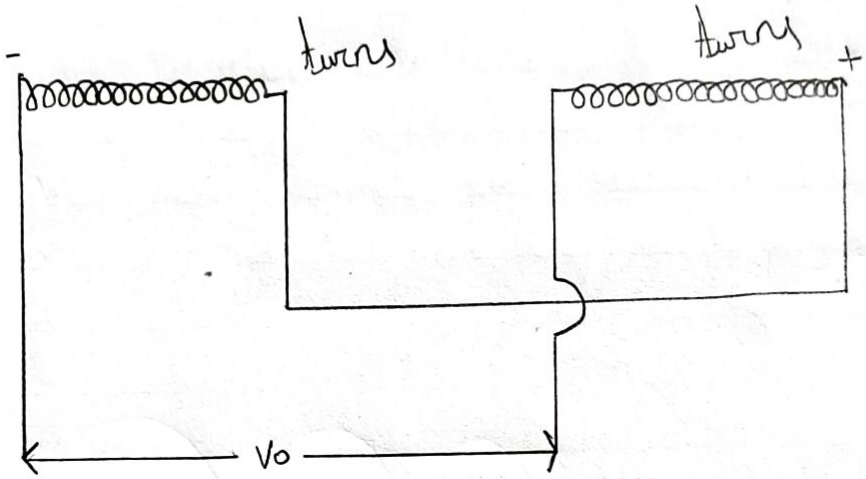
Laptop with internet connection

Theory

LVDT is an inductive transture that converts linear displacement into a electrical signal. It consist of a transformer having one primary winding and two secondary winding wound on a core. As the primary is connected to an A C source, the AC current and the voltage are produced in the secondary of LVDT. When the core is at the center the flux linking with both the secondary winding are equal when the core is at the right the flux linking with the secondary winding S1 is more so, the Emf induced in S1 is more than S2. so, the net Emf is positive when the core is at the left, the flux linking and the emf in S2 is more than S1. so, the net Emf is negative

CIRCUIT DIAGRAM





Tabulation

INPUT VOLTAGE		OUTPUT VOLTAGE	FREQUENCY	TIME = $\frac{1}{f}$
Positive Displacement	7	2.2V D=5mm	1000	$\frac{1}{1000}$ = 1ms
Negative Displacement	7	D=5mm	1000	$\frac{1}{1000}$ = 1ms

Expt. No. 6

procedure

* Connections are made as per the circuit diagram

* Sets the number of turns supplied voltage and supplied frequency

* Configure the parameters

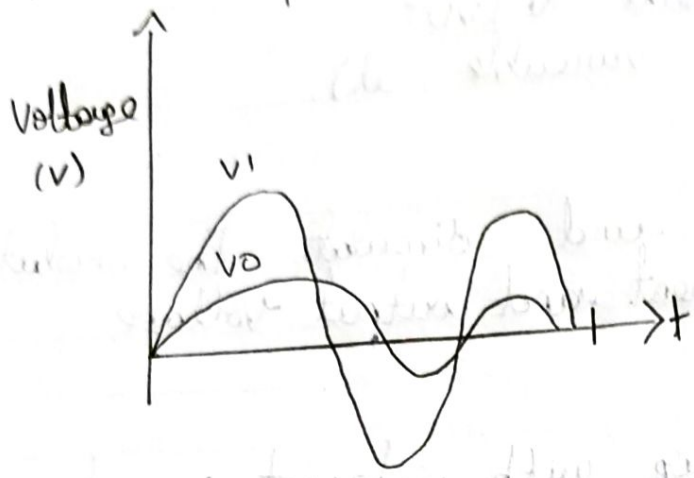
* Move the core to positive side and the negative side and observe the input voltage and output voltage wave forms

* Plot the graph between input and output voltage

SAVEETHA ENGINEERING COLLEGE SAVEETHA NAGAR 602 105 BE THE BEST 2001

Model Graph

Positive Displacement



Calculation

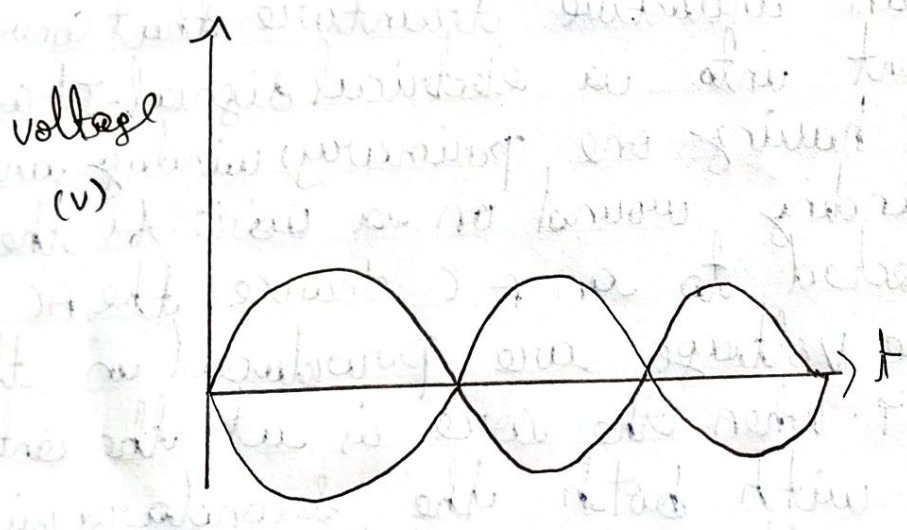
$V_m = \sqrt{2} \times V_{rms}$

$= \sqrt{2} \times 5$

$= 7V$

$T = 1ms$

Negative Displacement

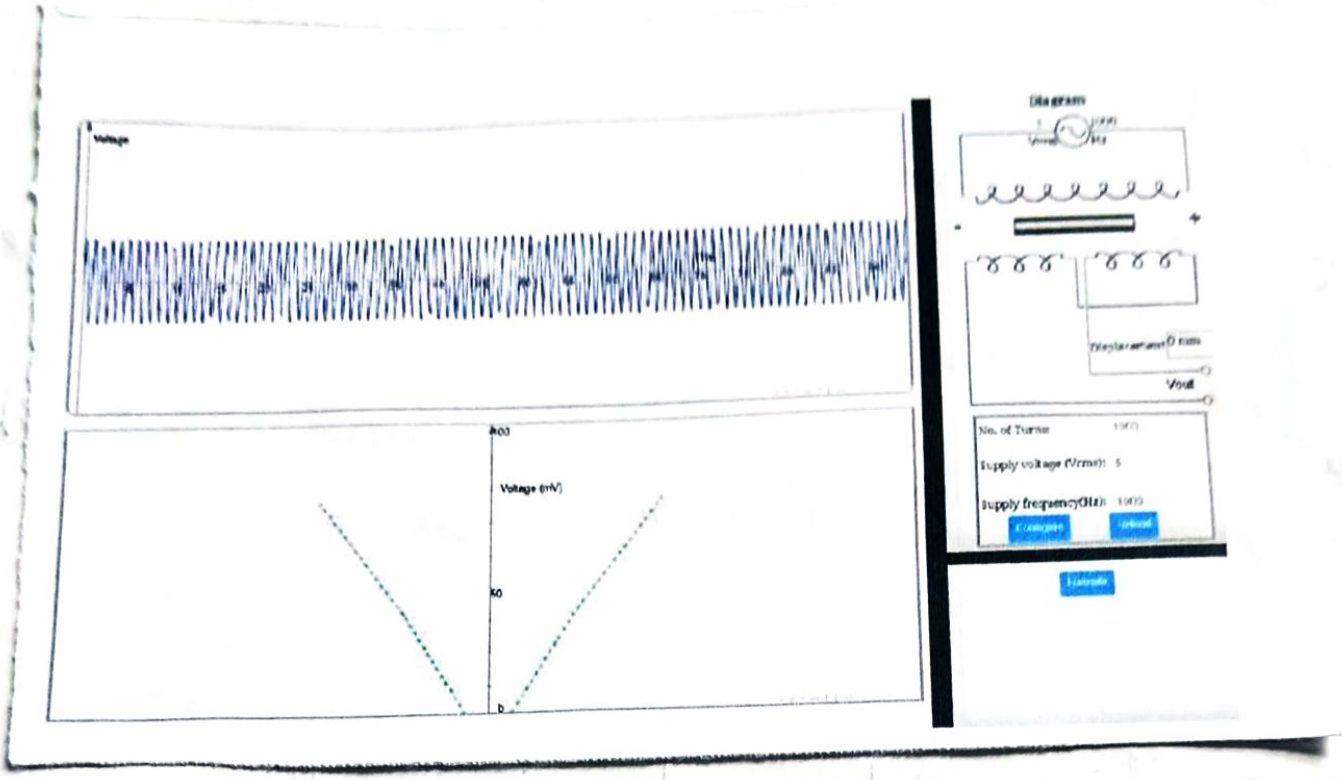


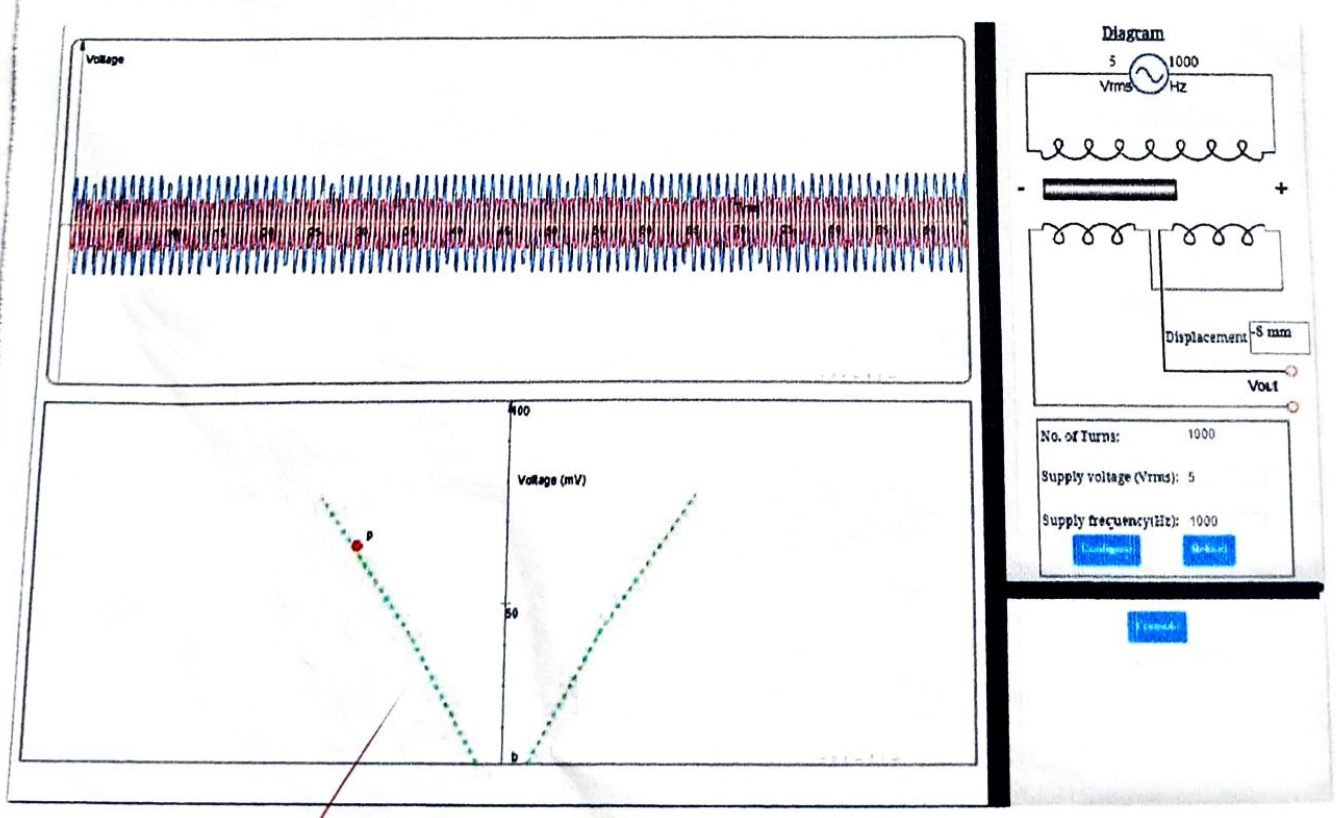
$4\ SD = 7$

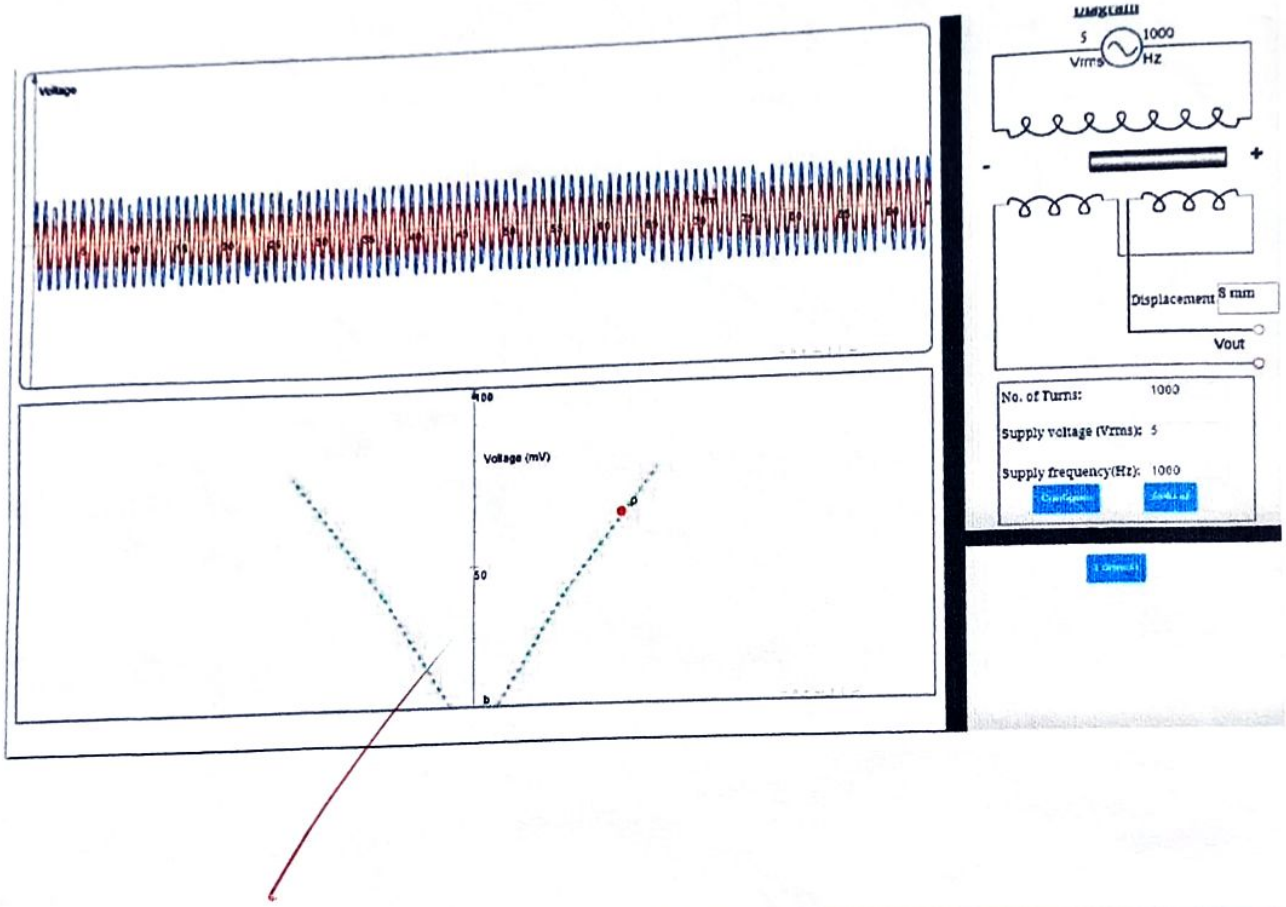
$SD = 7/4\ V$

$1.25SD = 7/4 \times 1.25$

$= 2.2V$


Voltage
Voltage (mV)
Diagram
Vout
No. of Turns
Supply voltage (Vrms): 5
Supply frequency(Hz): 1000


Voltage
Voltage (mV)
p
Diagram
5
1000
Vrms
Hz
Displacement -5 mm
Vout
No. of Turns: 1000
Supply voltage (Vrms): 5
Supply frequency(Hz): 1000


Voltage
400
Voltage (mV)
50
b
5
Vrms
1000
Hz
-
+
Displacement
5 mm
Vout
No. of Turns:
1000
Supply voltage (Vrms):
5
Supply frequency(Hz):
1000

Expt. No. 6

100

11/03/26

SAVEETHA
ENGINEERING COLLEGE
SAVEETHA NAGAR 602 105
BE THE BEST

Result :-

To understand and simulate the relation between core displacement and output voltage were developed and tested successfully.